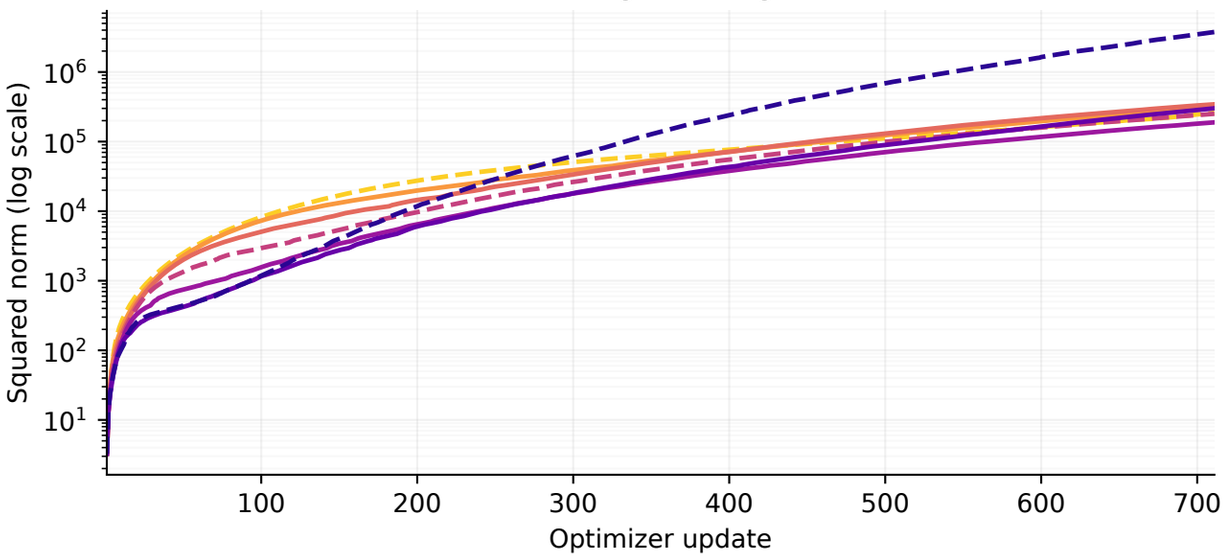
Pythia-14M: squared cumulative geometry across learning rates

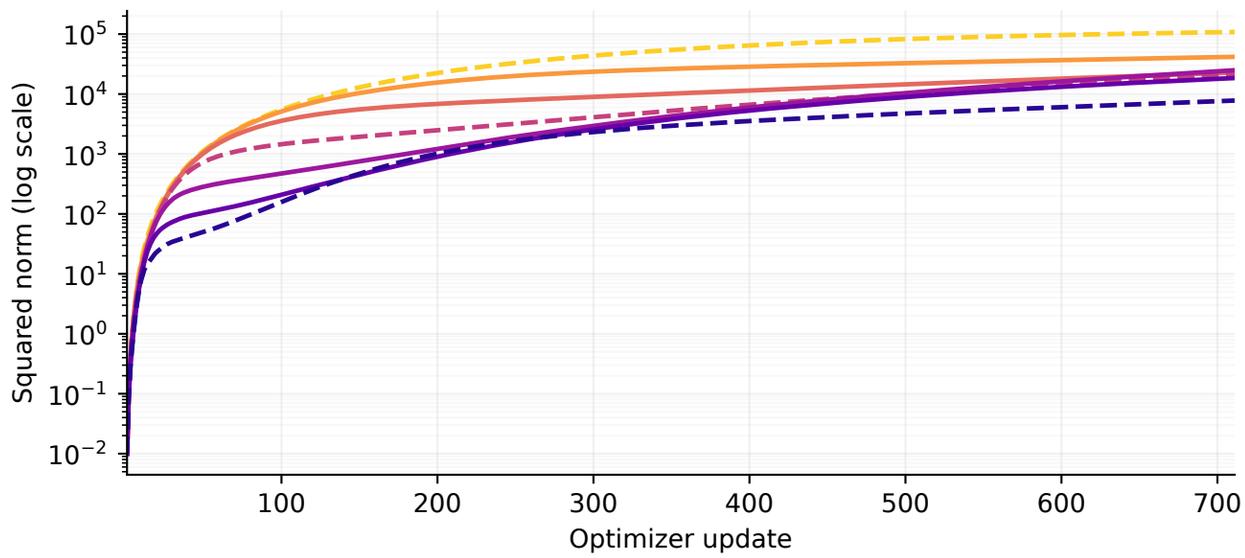
Peak LR: light yellow to dark purple as LR increases; S = Spark, R = RTX4090
Dashed = Spark; solid = RTX4090

0.0001 (S) · 0.000215443 (R) · 0.000464159 (R) · 0.001 (S) · 0.00215443 (R) · 0.00464159 (R) · 0.01 (S)

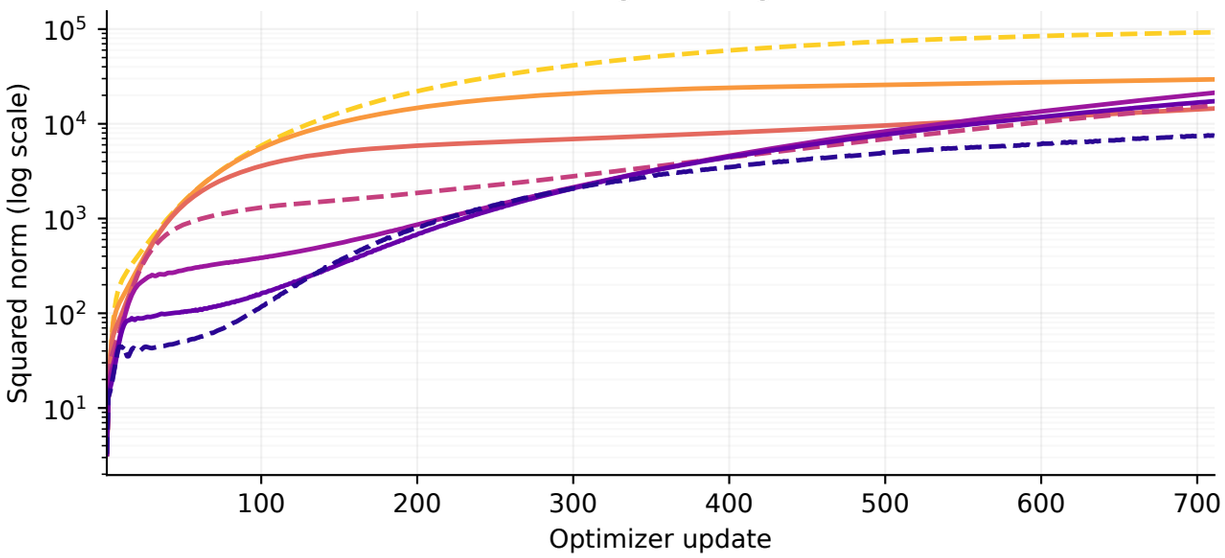
(a) Raw-gradient g: A^2



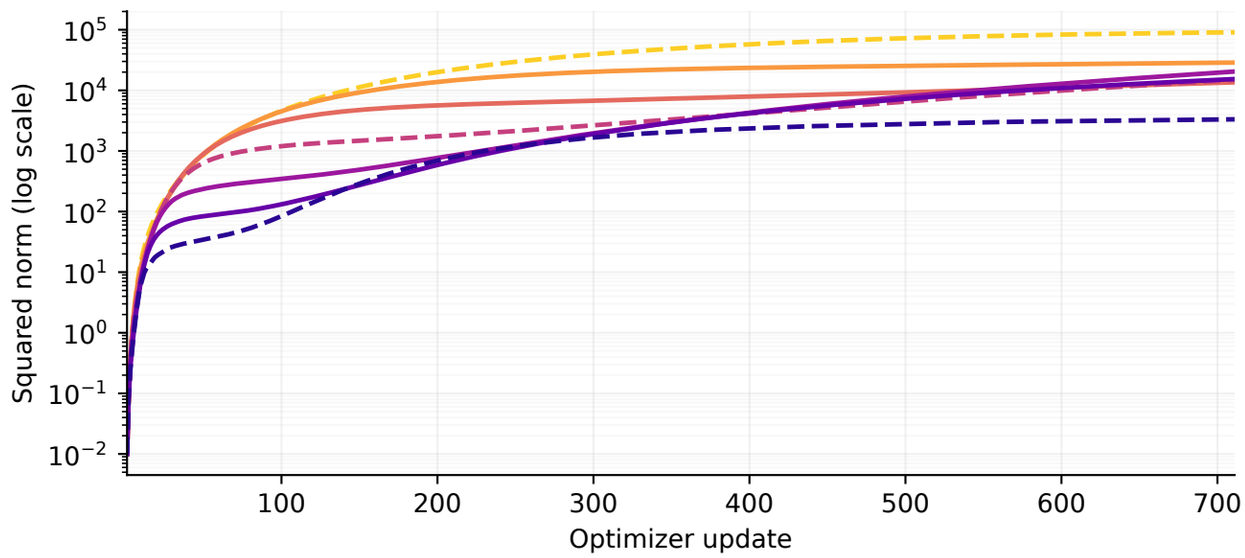
(b) Momentum m: A^2



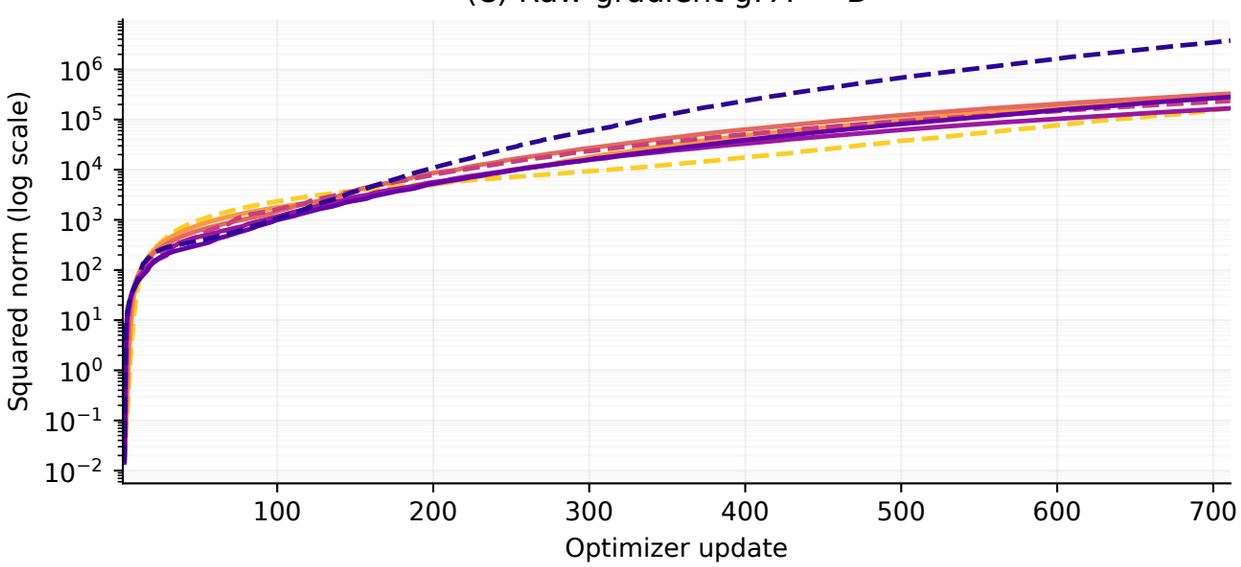
(c) Raw-gradient g: B^2



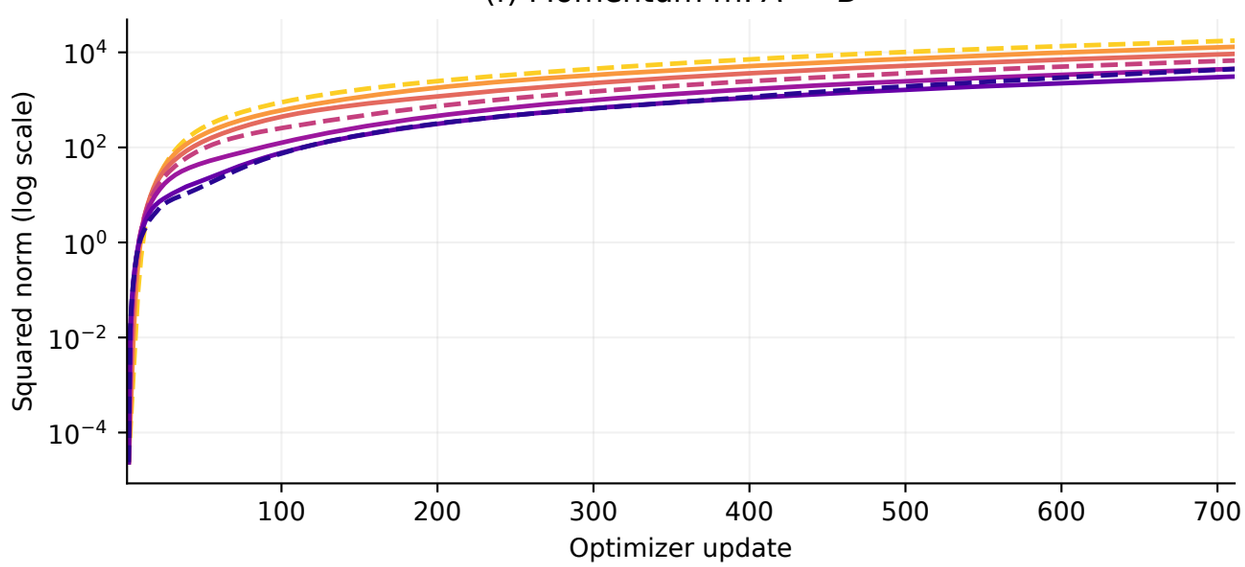
(d) Momentum m: B^2



(e) Raw-gradient g: $A^2 - B^2$



(f) Momentum m: $A^2 - B^2$



$A_t = (\sum_{s=1}^{t} \|v_s\|)$; $B_t = \|\sum_{s=1}^{t} v_s\|$. Rows: A_t^2, B_t^2, $A_t^2 - B_t^2$.
Unsmoothed curves; independent logarithmic Y scales; zero differences at update 1 are omitted from log display.
Columns: raw g before clipping; stored Adam momentum m before preconditioning.
Updates 1-711 shown; partial update 712 retained in results. Same frozen initialization/order; seed 1234.
S = Spark, R = RTX4090; hardware differs across groups. Momentum uses clipped gradients; LR also scales AdamW decay.